

$$1. \frac{1}{2} = 50\%$$

$$5. \frac{1}{6} = 16.66\%$$

$$9. \frac{1}{10} = 10\%$$

$$2. \frac{1}{3} = 33.33\%$$

$$6. \frac{1}{7} = 14.2857\%$$

$$10. \frac{1}{11} = 9.09\%$$

$$3. \frac{1}{4} = 25\%$$

$$7. \frac{1}{8} = \frac{1}{2} \left(\frac{1}{4} \right) = 12.5\%$$

$$11. \frac{1}{12} = \frac{1}{3} \left(\frac{1}{4} \right) = 8.33\%$$

$$4. \frac{1}{5} = 20\%$$

$$8. \frac{1}{9} = \frac{1}{3} \left(\frac{1}{3} \right) = 11.11\%$$

$$12. \frac{1}{16} = \frac{1}{2} \left(\frac{1}{8} \right) = \frac{1}{2} \times 12.5$$

$$13. \frac{1}{20} = 5\%$$

$$14. \frac{1}{25} = 4\%$$

$$= 6.25\%$$

Multiples:-

$$1. \frac{2}{3} \rightarrow 2 \times \left(\frac{1}{3} \right) = 2 \times 33.33\% = 66.66\%$$

$$6. \frac{5}{6} \rightarrow 5 \left(\frac{1}{6} \right) = 5 \times 16.66\%$$

$$2. \frac{3}{4} \rightarrow 3 \times \left(\frac{1}{4} \right) = 75\%$$

$$7. \frac{3}{8} \rightarrow 37.5\%$$

$$3. \frac{2}{5} \rightarrow 40\%$$

$$8. \frac{5}{8} \rightarrow 5 \times 12.5\% = 62.5\%$$

$$4. \frac{3}{5} \rightarrow 60\%$$

$$9. \frac{7}{8} \rightarrow 87.5\%$$

$$5. \frac{4}{5} \rightarrow 80\%$$

$$10. \frac{2}{9} \rightarrow 22.22\%$$

$$11. \frac{6}{9} \rightarrow 66.66\%$$

$$12. \frac{7}{9} \rightarrow 77.77\%$$

APTITUDE.

(Percentage)

1. If 20% of 45% of a number is 81. Find 40% of the number.

ANS: 360

$$\Rightarrow \frac{20}{100} \times \frac{45}{100} \times x = 81$$

$$\frac{40}{100} \times 900 = 360$$

$$x = 900$$

Easy: 20% of 45% 81

$$\begin{array}{ccc} \div 10 & & \div 10 \\ 2 & \times & 4.5 = 9\% \rightarrow 81 \end{array}$$

$$1\% \rightarrow 9$$

$$40\% \rightarrow 40 \times 9 = 360$$

2. A number is increased by $22\frac{1}{2}\%$ becomes 98.

Find the number.

ANS: 80

$$\Rightarrow 100\% + \frac{45}{2} = \frac{245}{2} \% \rightarrow 98$$

$$100\% \rightarrow x$$

$$98 \times \frac{2}{245} \times 100 = 80$$

$$\frac{245}{2} \times x = 98 \times 100$$

$$245x = 98 \times 200$$

$$x = \frac{98 \times 200}{245}$$

$$x = 80$$

3. If a number is increased by 216, it becomes 140% of itself. Find the number. ANS: 540

$$40\% \rightarrow 216$$

$$100\% \rightarrow x$$

$$\frac{40}{100}x = \frac{216}{100} \times 100$$

$$x = 540$$

$$40\% \rightarrow 216$$

$$100\% \rightarrow x$$

$$\frac{216}{40} \times 100 = 540$$

$$x + 216 = \frac{140}{100}x$$

$$x = 540$$

3. The sum of two numbers is 520. If the bigger is decreased by 4% and smaller number is increased by 12% then they become equal. Find the smaller number. ANS: 240

$$4\%$$

a (big)

$$12\%$$

b (small)

$$96\%a = 112\%b$$

$$\frac{a}{b} = \frac{112}{96}$$

$$\frac{a}{b} = \frac{7}{6}$$

$$7 \text{ parts} \rightarrow 520$$

$$(b) \quad 6 \text{ parts} \rightarrow b$$

$$b = 6 \times 40 = 240$$

$$96\%a = 112\%b$$

$$\frac{a}{b} = \frac{112}{96}$$

$$\frac{a}{b} = \frac{7}{6} \Rightarrow a = \frac{7}{6}b$$

$$a + b = 520$$

$$\frac{7b}{6} + b = 520$$

$$b = 240$$

4. The profit made by a company one year is Rs. 150000. Two years ago the profit made by the same company is Rs. 240000. What is the percentage change in the profit made by company.

a) 66.66% increase

b) 66.66% dec

ANS: d) 37.5% dec.

b) 37.5% inc

d) 37.5% dec.

$$\% \text{ inc} = \frac{\text{difference}}{\text{small no}} \times 100\%$$

$$\% \text{ dec} = \frac{\text{difference}}{\text{big no}} \times 100\%$$

2 years ago

Now

Rs. 240000

Rs. 150000

dec

$$\% \text{ dec} = \frac{90000}{240000} \times 100\%$$

$$= 37.5\% \text{ dec.}$$

5. In a test consists of 300 ques, a student answered 40% among first 100 questions correctly. What % of remaining 200 questions, he need to answer correctly, so that he can get 50% over entire exam.

ANS: 55%

50% of 300 is 150

need to answer correctly

40% of 100 is 40

in 1st 100 goes 40 correct

$$100 \rightarrow 40$$

$$200 \rightarrow 110$$

$$\% = \frac{110}{200} \times 100\%$$

$$= 55\%$$

6. A vendor sells 50% of the apples he had and throws 20% of the remaining apples. Next day he sells 60% of the remaining apple and throws away the rest. what percentage of apple did the vendor thrown away?

ANS: 26%

Total Apple = 100

Day - 1

$$\text{sold} \rightarrow 50$$

$$\text{throw} \rightarrow \frac{20}{100} \times 50$$

10

Day - 2 (40)

$$\text{sold} \rightarrow \frac{60}{100} \times 40 = 24$$

$$\text{throw} \rightarrow 16$$

total % throw = 26%

7. In a triangle, base is increased by $66\frac{2}{3}\%$ then by how much percentage height should be decreased, so that area of triangle remains constant.

ANS: 40%

$$66\frac{2}{3}\% \rightarrow \frac{2}{3} \times 100 = 66.66\%$$

$$\text{ breadth } b = 100\% + 66\frac{2}{3}\% = 1 + \frac{2}{3} = \frac{5}{3}$$

$$b' = \frac{5}{3}b$$

$$A \Rightarrow \frac{1}{2} \times b \times h$$

$$h \rightarrow \frac{3}{5}h$$

$$A' \Rightarrow \frac{1}{2} \times \frac{5}{3}b \times \frac{3}{5}h$$

$$\% \text{ dec} = \frac{h - \frac{3}{5}h}{h} \times 100\%$$

$$= \frac{2h}{5h} \times 100\%$$

$$= 40\%$$

8. Price of a commodity is increased by 24% and the quantity purchased is decreased by 20%. What is the percentage increase or decrease in the amount spent on that commodity.

ANS: 0.8% decrease.

Expenditure
Price Quantity

$$E = P \times Q$$

Shortcut: inc
1 dec.

$$\frac{E_1}{E_2} = \frac{P_1}{P_2} \times \frac{Q_1}{Q_2}$$

$$a - b - \frac{ab}{100}$$

$$\frac{E}{E_2} = \frac{100}{124} \times \frac{100}{80} = \frac{125}{124}$$

$$a = 24 \quad b = 20$$

$$E_2 = \frac{124}{125} E$$

$$24 - 20 - \frac{20 \times 24}{100}$$

$$4 - 4.8 = -0.8$$

0.8 decreases.

$$= \frac{124}{125} \times 100$$

$$E_2 = 99.2\%$$

$$\Rightarrow 100 - 99.2\% \Rightarrow 0.8\%$$

9. Initial length of a rectangular box is 20 cm.

The box is remade such that its length is increased by 30% but its breadth is reduced by 20%. After this its area is increased by 100 cm^2 . Find the new area of the box.

ANS: $A' = 2600 \text{ cm}^2$

case 1: $l = 20 \text{ cm}$ b

case 2: $20 + \frac{30}{100} \times 20$

$l' = 26 \text{ cm}$

$b' = b - \frac{b}{5}$ $\frac{20}{100} \times b = \frac{b}{5}$

$b' = \frac{4b}{5}$

$A' = l' \times b'$

$A' = A + 100$

$\frac{4b}{5} = 100$

$b' = \frac{4 \times 125}{5}$

$b' = 100$

$26 \times \frac{4b}{5} = 20b + 100$

$b = 125 \text{ cm}$

$\frac{104b}{5} - 20b = 100$

$A' = 26 \times 100 = 2600 \text{ cm}^2$

10. A man's annual income has increased by 1.2 lakhs. But tax on the income he has to pay is reduced from 12% to 10%. He now pays same amount of tax as before. Find his new salary.

ANS: 7.2 Lakhs

Tax = Income $\times$ Tax%

$I_1 T_1 = I_2 T_2$

Initial income = Rs. x Lakhs

Final income = $x + 1.2$ Lakhs

$\frac{I_1}{I_2} = \frac{T_2}{T_1} \Rightarrow \frac{x}{x+1.2} = \frac{10}{12}$

$= 6.1.2$

$= 7.2$

$12x = 10x + 12$

$2x = 12$

$x = 6$

11. Price of a movie ticket in a hall is decreased by 1%, but the total revenue collected is increased by $83\frac{1}{3}\%$. Because seating capacity was increased. If the final seating capacity now is 350, find the initial seating capacity of the hall.

ANS: 189 seats

Revenue = price $\times$ seating capacity.

$$\frac{R_1}{R_2} = \frac{P_1}{P_2} \times \frac{S_1}{S_2}$$

$$\frac{100}{\left(\frac{550}{3}\right)} = \frac{100}{9} \times \frac{S_1}{350}$$

$$\left[100 + \frac{250}{3} = \frac{550}{3} \right]$$

$$S_1 = 100 \times \frac{3}{550} \times \frac{99}{100} \times 350$$

$$S_1 = 189 \text{ seats}$$

12. A village lost 12% of its sheep in a flood, and 5% of the remaining died from diseases. If the no. of sheep left now is 8360. Find the original no. of sheep in the village before flood.

ANS: 10000

$$\Rightarrow a + b + \frac{ab}{100}$$

$$\Rightarrow 12 + 5 + \frac{60}{100}$$

$$12\% \text{ of } 5\% = 8360$$

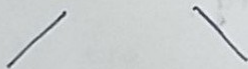
$$\frac{88}{100} \times \frac{95}{100} \times x = 8360$$

$$x = 10000$$

13. The cost of an apple is twice that of a banana. The cost of a banana is 25% less than pineapple. The cost of each type of fruit is increased by 10%. Then find the percentage increase in cost of 4 banana, 2 apples and 3 pineapples.

ANS: 10%

$$4B + 2A + 3P$$



$$4 \times 75 + 2 \times 150 + 3 \times 100$$

$$= 900/-$$

$$4 \times 82.5 + 2 \times 165 + 3 \times 110$$

$$= 990/-$$

$$\% \text{ Inc} = \frac{990 - 900}{900} \times 100 = \frac{90}{900} \times 100 = 10\%$$

14. A person spends 16.66% of his income on house rent, 40% of remaining on food and clothes and 10% of remaining on education. He deposited the remaining amount in bank. Find the ratio of total income to total expenditure.

ANS: 20:11

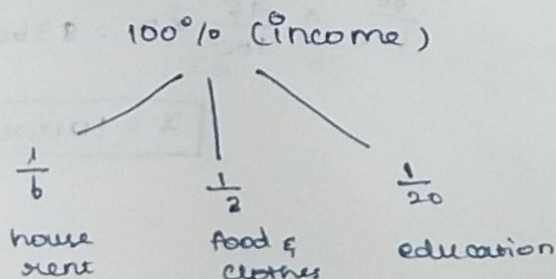
$$16.66\% \rightarrow \frac{1}{6}$$

$$\text{rem } \frac{5}{6}$$

$$\frac{40}{100} \times \frac{5}{6} = \frac{1}{3}$$

$$1 - \left[\frac{1}{6} + \frac{1}{3} \right]$$

$$1 - \frac{1}{2} = \frac{1}{2}$$



$$E = \frac{1}{6} + \frac{1}{3} + \frac{1}{20} = \frac{11}{20}$$

$$I:E = 1:\frac{11}{20} \quad (\times 20)$$

$$= 20:11$$

15. In an examination a student must score 40% marks to pass. A student who gets 220 marks fails by 20 marks. Find the maximum marks in the examination.

ANS: 600

$$40\% = 220 + 20$$

$$40\% = 240$$

$$100\% = x$$

$$40x = 240 \times 100$$

$$x = 600$$

16. If a student scores 43% of marks, then he will be failed by 72 marks but when he scores 57% of marks, he will be passed by 40 marks. Find the pass mark percentage.

ANS: 52%

$$\text{Total marks} = x$$

$$\text{Pass mark} = y$$

$$\% \text{ pass} = \frac{y}{x} \times 100\%$$

$$\frac{43}{100} \times x = y - 72$$

$$\frac{14x}{100} = (y + 40) - (y - 72)$$

$$\frac{57}{100} \times x = y + 40$$

$$x = 800$$

Total marks

$$\Rightarrow \frac{43}{100} \times 800 = y - 72$$

$$\boxed{y = 416}$$

$$\% \text{ pass} = \frac{416}{800} \times 100\%$$

$$= 52\%$$

17. In an election between two candidates, winner secured 58% of total votes and wins by a majority of 2400 votes. How many votes did the loser get.

$$\boxed{\text{ANS: } 6300}$$

$$W \rightarrow 58\%$$

$$L \rightarrow 42\%$$

$$58\% - 42\% = 2400$$

$$16\% = 2400$$

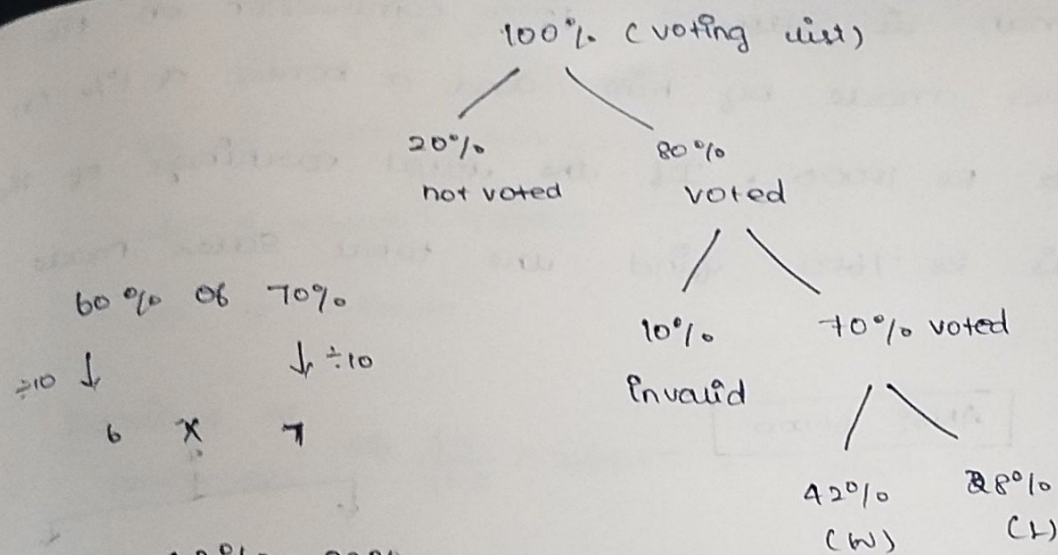
$$\frac{8}{16}x = \frac{300}{2400} \times \frac{21}{42}$$

$$42\% = x$$

$$\boxed{x = 6300}$$

18. In an election, 20% of the voters did not vote, $12\frac{1}{2}\%$ of votes are declared invalid, and the winner gets 60% of valid votes and won the election by 4200 votes. Find the no. of voters in the voting list.

$$\boxed{\text{ANS: } 30000}$$



$$42\% - 28\% \Rightarrow 14\%$$

$$14\% \rightarrow 4200$$

$$1\% \rightarrow 300$$

$$100\% \rightarrow 30000$$

19. In a rectangle $l = 3b$, if the length and breadth of rectangle is increased by 30% and 10% .
 find the percentage increase in the perimeter of rectangle .

ANS: 25%

$$l = 3b$$

$$P_1 = 2(3b + b)$$

$$P_1 = 8b$$

$$P_2 = 2(3.9b + 1.1b) = 10b$$

$$l' = 130\% l = \frac{130}{100} \times 3b = 3.9b$$

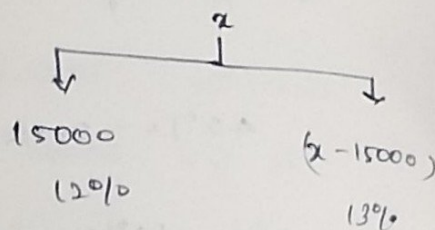
$$b' = 110\% b = \frac{110}{100} \times b = 1.1b$$

$$\% \text{ inc} = \frac{2b}{8b} \times 100 = 25\%$$

20. A salesman is allowed 12% commission on the total sales made by him and a bonus of 1% on the sales over Rs. 15000. If the total earnings of the salesman is Rs. 7650 find the total sales made by him.

ANS: 60000

$$\frac{12}{100} \times 15000 + \frac{13}{100} \times (x - 15000) = 7650$$



$$\frac{13x}{100} - \frac{1}{100} \times 15000 = 7650$$

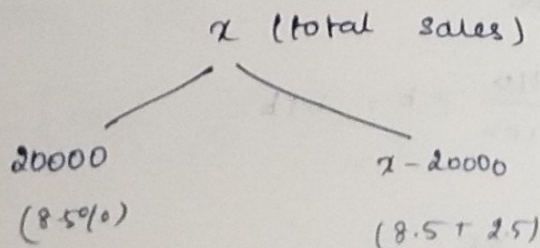
$$\frac{13x}{100} = 7800$$

$x = 60000$

21. A salesman is allowed 8.5% commission on total sales and 2.5% bonus on the sales over Rs. 20000. If the bonus he received is Rs. 2250. Find the earning of salesman including bonus.

ANS:

$$\frac{8.5}{100} \times 20000 + \frac{2.5}{100} \times (x - 20000) = 2250$$



$$\frac{2.5}{100} \times (x - 20000) = 2250$$

$$\Rightarrow x - 20000 = 40 \times 2250$$

$$\Rightarrow x - 20000 = 90000$$

$$\Rightarrow x = 110000$$

Earnings of
salesman

$$\Rightarrow \frac{8.5}{100} \times 20000 + 90000 \times \frac{11}{100}$$

$$\Rightarrow 1700 + 9900 = 11600$$